

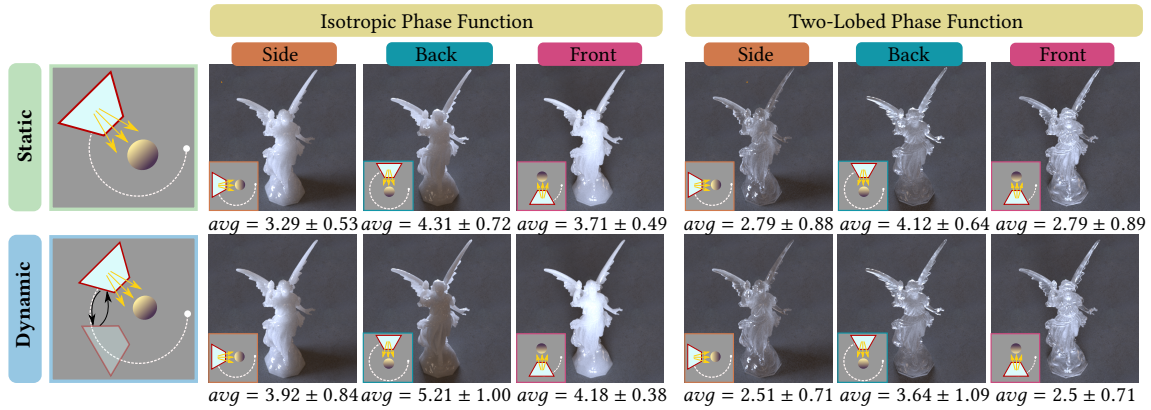
# On the Influence of Dynamic Illumination in the Perception of Translucency

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**Figure 1:** We analyze the effect of dynamic natural illumination on the perception of translucent materials. We design a matching experiment where participants estimate the optical density (the extinction coefficient  $\sigma_t$ ) of a reference object with *static* (top row) or *dynamic* (bottom row) illumination. We do this under three different light directions in the match stimulus (*side*, *back* and *front*, depicted in the inset of each image), for different optical properties of the translucent medium. The images depict our test object, rendered with the average density estimated by participants for each condition, for a fixed reference density  $\sigma_t = 4.0$  (below we report mean and standard deviation of the estimated density).

## ABSTRACT

Translucent materials are ubiquitous in the real world, from organic materials such as food or human skin, to synthetic materials like plastic or rubber. While multiple models for translucent materials exist, understanding how we perceive translucent appearance, and how it is affected by illumination and geometry, remains an open problem. In this work, we analyze how well human observers estimate the density of translucent objects for static and dynamic illumination scenarios. Interestingly, our results suggest that dynamic illumination may not be critical to assess the nature of translucent materials.

## KEYWORDS

Lighting; material perception; translucency.

### ACM Reference Format:

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## 1 INTRODUCTION

We make judgements on material appearance on a daily basis, as we interact with the real world. Translucent materials are ubiquitous in the real world and can exhibit a wide range of appearances. While physical simulations to model the behavior of these materials are well understood, the way in which the physical parameters affect the perceived appearance of translucent materials is still an open field of research. Working towards this goal, researchers have shown how the perception of translucency can be influenced by, e.g., the position of the light, or the material density and phase function [Fleming and Bülthoff 2005; Gkioulekas et al. 2013; Xiao et al.

2014]. Other studies have focused on investigating the influence of translucency over perceived glossiness [Gigilashvili et al. 2021]. We build on top of these studies, and extend them to a dynamic setting. Previous works have shown that dynamic stimuli can improve appearance perception, such as glossiness constancy [Doerschner et al. 2011]. Inspired by these findings, we design an experiment in which we evaluate *whether having dynamic illumination improves the perception of translucency as compared to the static case*. We do this using a matching task, in which participants match the optical density (extinction coefficient  $\mu_t$ ) of a match image to that of a reference stimulus. This is done for a variety of light positions, phase functions and density levels.

Interestingly, our results suggest that a dynamic illumination does not lead to an improvement in the perception of the density of translucent materials, and a simpler, static illumination can suffice to convey this aspect of the material.

## 2 METHOD

Our study is based on an asymmetric matching task, where each participant has to match the optical density of the match image to that of the reference stimulus [Xiao et al. 2014]. Reference and match stimuli are shown side by side, and show the same object (the *Lucy* model). They are both rendered with the same optical properties (surface roughness, single scattering albedo, and phase function), except for the optical density (extinction coefficient). Based on the findings by Gkioulekas et al. [2013] we fix the roughness and single scattering albedo and use two different phase functions (isotropic, and a forward two-lobed phase function, see Fig. 1). The matching task is done for four different optical density levels. In all cases, the scene is lit using the *Ennis* environment map, but rotated so that the direction of maximum intensity hits the object differently. In particular, the match image can be lit in three different static conditions: *side*, *back* and *front* (see Fig. 1).

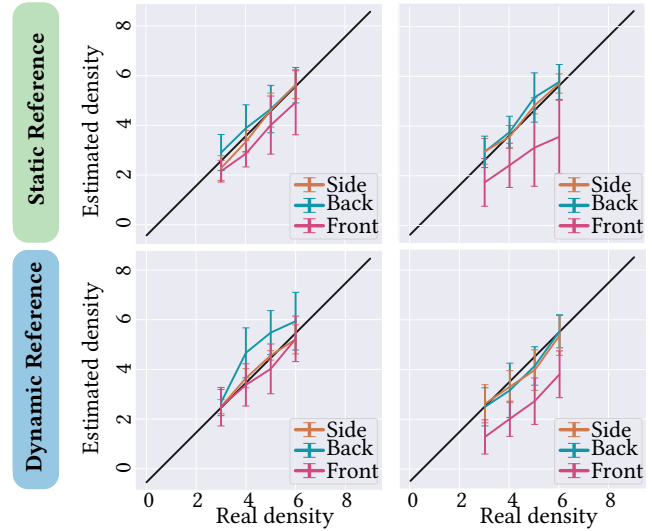
Twelve participants with normal or corrected-to-normal vision perform two different matching experiments, in random order and in two separate sessions: One with a static reference, and one with an animated dynamic illumination. In the case of static illumination, the object is lit from a *side* lighting condition, at about 30° from the match’s *side* condition, to avoid pixel-wise matching. In the dynamic experiment, the illumination rotates back and forth from a near back-lighting condition to a front-lighting one.

In total, each of the two experiments is composed by 18 matching pairs, shown in random order: three *match light positions*, two *phase functions* (an isotropic and a double-lobed forward-scattering phase function), and four *density levels*, ranging from very thin semi-transparent media to very optically-thick media.

## 3 ANALYSIS AND RESULTS

We analyze participants’ performance when estimating the density of translucent objects by comparing the estimated optical density (extinction coefficient  $\mu_t$ ) of the match image against that of the reference stimulus. Fig. 2 shows the mean estimated density as a function of the ground truth density for the different conditions tested. We analyze our data using a repeated measures ANOVA.

We find a significant effect of the match light position ( $p \leq 0.05$ ): Depending on the lighting position, participants either overestimate



**Figure 2: Mean estimated density with respect to the ground-truth reference density for each match light position (*side*, *back* and *front*), for an isotropic phase function (left) and a two-lobed phase function (right), and for the static (top row) and dynamic (bottom row) illumination experiments.**

or underestimate the density (see Fig. 2). This is consistent with previous work [Xiao et al. 2014]; further, Xiao et al. used lower-frequency synthetic illumination, and we show that the effect holds with more realistic environment maps.

Surprisingly, we find no statistically significant differences ( $p > 0.05$ ) between the static and dynamic illumination of the reference. Density estimation is similar between both conditions, as illustrated by the graphs in Fig. 2. This is an unexpected result, since dynamic lighting conditions offer more information (i.e., light incoming directions), and previous work had shown an improvement in appearance perception under motion for other material properties [Doerschner et al. 2011]. A possible explanation is that users are not able to leverage the extra information provided by the dynamic motion of an uncontrolled light source. Still, further research is needed to fully understand which factors affect translucency perception. In particular, it can be interesting try similar experiments including more shapes, light configurations, and scattering parameters (e.g. roughness, back-scattering phase functions, etc.).

## REFERENCES

- Katja Doerschner, Roland W Fleming, Ozgur Yilmaz, Paul R Schrater, Bruce Hartung, and Daniel Kersten. 2011. Visual motion and the perception of surface material. *Current Biology* 21, 23 (2011), 2010–2016.
- R. W. Fleming and H. H. Bühlhoff. 2005. Low-Level Image Cues in the Perception of Translucent Materials. *ACM Trans. Appl. Percept.* 2, 3 (2005), 346–382.
- Davit Gigilashvili, Weiqi Shi, Zeyu Wang, Marius Pedersen, Jon Yngve Hardeberg, and Holly Rushmeier. 2021. The Role of Subsurface Scattering in Glossiness Perception. *ACM Trans. Appl. Percept.* 18, 3 (2021), 1–26.
- Ioannis Gkioulekas, Bei Xiao, Shuang Zhao, Edward H. Adelson, Todd Zickler, and Kavita Bala. 2013. Understanding the Role of Phase Function in Translucent Appearance. *ACM Trans. Graph.* 32, 5 (2013), 147:1–19.
- Bei Xiao, Bruce Walter, Ioannis Gkioulekas, Todd Zickler, Edward Adelson, and Kavita Bala. 2014. Looking against the light: How perception of translucency depends on lighting direction. *Journal of Vision* 14, 3 (2014), 1–17.